VALUATIVE EXISTENCE FOR THE QUOT FUNCTOR

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 769-776 of sources/stacks-moduli/pinned-original-source.tex; proof span is lines 778-825, the complete author proof (46 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 6 of queue_moduli.json (pb=2311, stmt 769-776, proof 778-825).

Editorial note. stacks-moduli wave-1: fresh adjudication accept (2026-10-14); pool per Chief order 08:5x (dpa superseded to lane-s9); moduli.tex 53 labels/48 with proof judged 25 accept / 23 reject / 5 pb=0 env-dup rows excluded; spans cite-free and reference/slogan/history-free so no patchenv needed.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0539: sources/stacks-moduli/pinned-original-source.tex, source lines 825–825. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DLR: sources/stacks-moduli/pinned-original-source.tex, source lines 825–825. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03IX: sources/stacks-moduli/pinned-original-source.tex, source lines 825–825. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03M3: sources/stacks-moduli/pinned-original-source.tex, source lines 825–825. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Assume $X \to B$ is proper as well as of finite presentation and $\mathcal{F}$ quasi-coherent of finite type. Then $\mathrm{Quot}_{\mathcal{F}/X/B} \to B$ satisfies the existence part of the valuative criterion (Morphisms of Spaces, Definition 03IX).*

Date: stacks-project:moduli.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Taking base change, this immediately reduces to the following problem: given a valuation ring R with fraction field K, an algebraic space X proper over R, a finite type quasi-coherent $\mathcal{O}_X$-module $\mathcal{F}$, and a coherent quotient $\mathcal{F}_K \to \mathcal{Q}_K$, show there exists a quotient $\mathcal{F} \to \mathcal{Q}$ where $\mathcal{Q}$ is a finitely presented $\mathcal{O}_X$-module flat over R whose generic fibre is $\mathcal{Q}_K$. Observe that by Flatness on Spaces, Theorem 0DLR any finite type quasi-coherent $\mathcal{O}_X$-module $\mathcal{F}$ flat over R is of finite presentation. We first solve the existence of $\mathcal{Q}$ affine locally.

Affine locally we arrive at the following problem: let $R \to A$ be a finitely presented ring map, let M be a finite A-module, let $\varphi : M_K \to N_K$ be an A_K-quotient module. Then we may consider

$$L = \{x \in M \mid \varphi(x \otimes 1) = 0\}$$

The $M \to M/L$ is an A-module quotient which is torsion free as an R-module. Hence it is flat as an R-module (More on Algebra, Lemma 0539). Since M is finite as an A-module so is L and we conclude that L is of finite presentation as an A-module (by the reference above). Clearly M/L is the unique such quotient with $(M/L)_K = N_K$.

The uniqueness in the construction of the previous paragraph guarantees these quotients glue and give the desired $\mathcal{Q}$. Here is a bit more detail. Choose a surjective étale morphism $U \to X$ where U is an affine scheme. Use the above construction to construct a quotient $\mathcal{F}|_U \to \mathcal{Q}_U$ which is quasi-coherent, is flat over R, and recovers $\mathcal{Q}_K|U$ on the generic fibre. Since X is separated, we see that $U \times_X U$ is an affine scheme étale over X as well. Then $\mathcal{F}|_{U \times_X U} \to \mathrm{pr}_1^*\mathcal{Q}_U$ and $\mathcal{F}|_{U \times_X U} \to \mathrm{pr}_2^*\mathcal{Q}_U$ agree as quotients by the uniquess in the construction. Hence we may descend $\mathcal{F}|_U \to \mathcal{Q}_U$ to a surjection $\mathcal{F} \to \mathcal{Q}$ as desired (Properties of Spaces, Proposition 03M3). $\square$